

The Heart

Two pumps in one organ, wired to fire itself. Everything it does comes down to pressure.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Trace a drop of blood through both circuits, in order
- Explain what opens and closes a heart valve
- Follow the wiring, and say why the pause at the AV node matters
- Match the parts of an ECG to what is happening electrically
- Calculate cardiac output, and predict what raises or lowers it

1 Two circuits, one organ

The right heart sends blood to the lungs. The left heart sends it everywhere else.

Body → right atrium → right ventricle → lungs → left atrium → left ventricle → body.

Blood low in oxygen is on the right. Blood full of oxygen is on the left. Both sides pump the same amount every beat.

They do not do the same amount of work. The right ventricle pushes blood a short way into a low-pressure circuit. The left pushes it around the whole body.

That is why the left wall is several times thicker. Same volume, far more pressure.

WHY THE NURSE CARES

Which side fails tells you where fluid backs up. Left-sided failure backs up into the lungs, so the patient is short of breath. Right-sided failure backs up into the body, so the legs swell and the neck veins stand out.

2 Valves are not muscles

No valve ever decides to open. Pressure opens it and pressure slams it shut.

Every valve is a one-way flap. It swings whichever way the pressure difference pushes it. Higher pressure behind it, and it opens. Higher pressure in front, and it shuts.

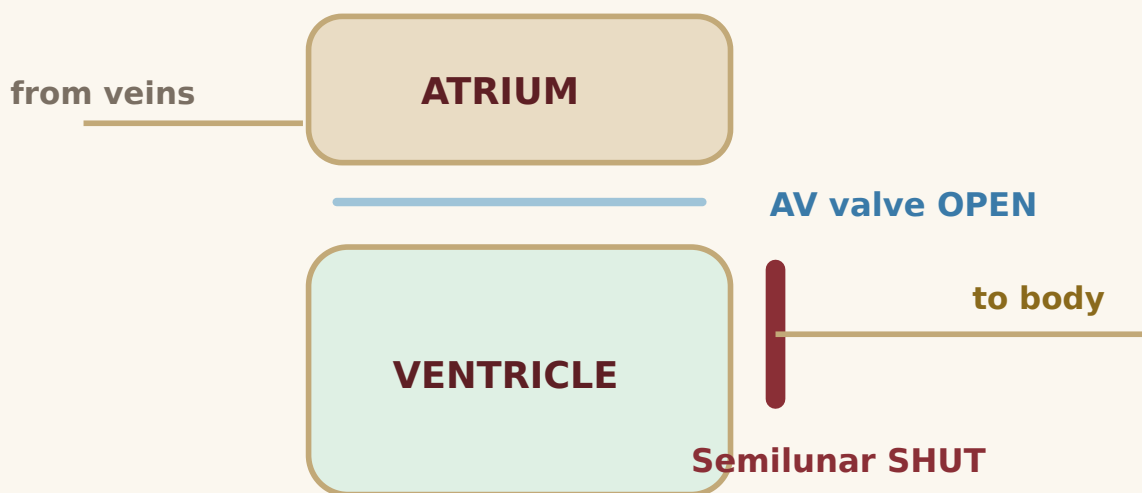
The **heart sounds** are those closures. The first sound is the upper valves shutting as the ventricles start to squeeze. The second is the outlet valves shutting as the ventricles relax.

THINK OF IT THIS WAY

A swinging door in a drafty hallway. Nobody operates it. Whichever side has more pressure decides which way it sits.

3 One beat, step by step

WALK THROUGH THE CARDIAC CYCLE



Ventricles are relaxed and their pressure is low, so the AV valve hangs open and blood pours in from the atrium on its own. Around 70–80% of filling happens here, before the atrium has done anything at all.

Two things are worth taking from that sequence.

First, most filling is **passive**. Blood pours through the open valve before the atrium contracts at all, and the atrial squeeze only tops it up.

Second, the ventricle spends a moment squeezing with **every valve shut**. It builds pressure without pushing any blood out, because it has to beat the pressure in the artery before the door will open.

4 The heart's own wiring

- 1 **The SA node fires on its own**, about 60 to 100 times a minute. It is the pacemaker because it is the fastest.
- 2 **The atria depolarize** and contract.
- 3 **The AV node holds the signal briefly**. That pause lets the atria finish emptying before the ventricles start.
- 4 **Fast fibers carry the signal** through the ventricles.
- 5 **The ventricles contract** from the bottom up, which pushes blood toward the outlet valves.

ECG PART	WHAT IS HAPPENING
P wave	The atria are about to contract
QRS complex	The ventricles are about to contract
T wave	The ventricles are resetting

An ECG records *electrical* activity only. It tells you the signal was sent. It does not prove the muscle answered.

That difference is why a patient can have an organized rhythm on the monitor and no pulse at all.

The nerves do not create the beat. They edit it. Sympathetic input speeds the pacemaker and increases force. The vagus nerve slows it. Take away all nerve input and the heart keeps beating at its own rate, which is exactly what happens after a transplant.

5 Cardiac output — the number everything serves

Cardiac output = heart rate × stroke volume. About 70 beats a minute times about 70 mL is roughly 5 liters a minute.

Stroke volume depends on three things:

- **Preload** is how full the ventricle is before it squeezes. More filling stretches the muscle, and it contracts harder. This is the same length-and-force idea from Module 7, applied to the heart.
- **Contractility** is how hard the muscle squeezes for a given stretch. Sympathetic drive and some drugs raise it.
- **Afterload** is the pressure the ventricle has to beat to open the outlet valve. Higher afterload means less blood leaves.

Watch how these work together when a patient bleeds. Volume falls, so preload falls, so stroke volume falls.

The body makes up for it by speeding the heart and narrowing the vessels. That is why a fast heart and cool, pale skin show up before the blood pressure ever drops.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. Why is the left ventricle wall thicker than the right?

- a) It pumps more blood each beat
- b) It has to make much higher pressure to supply the whole body
- c) It has more valves
- d) It receives blood from two atria

2. What makes a heart valve close?

- a) Muscle inside the valve contracts
- b) Higher pressure on the far side pushes it shut
- c) A nerve signal closes it
- d) It closes when blood cools

3. Why does the AV node pause before passing the signal on?

- a) To let the atria finish emptying before the ventricles squeeze
- b) To slow the overall heart rate
- c) To let the heart's own arteries fill
- d) To let the SA node recover

4. A monitor shows an organized rhythm but the patient has no pulse. What does that tell you?

- a) The monitor is broken
- b) The electrical signal is there, but the muscle is not pumping effectively
- c) The heart has stopped conducting
- d) The blood pressure is fine

5. A patient is bleeding. Which change shows up first?

- a) Blood pressure drops sharply
- b) Heart rate rises and vessels narrow to compensate
- c) Stroke volume rises
- d) Cardiac output rises above normal

Answer key

1. It has to make much higher pressure to supply the whole body

Correct. Same volume, much greater pressure, so much more muscle.

2. Higher pressure on the far side pushes it shut

Right. Every valve is passive. Pressure alone decides which way it sits.

3. To let the atria finish emptying before the ventricles squeeze

Correct. Without the pause, the chambers would squeeze together and filling would be lost.

4. The electrical signal is there, but the muscle is not pumping effectively

Correct. An ECG shows the signal being sent, not the muscle answering.

5. Heart rate rises and vessels narrow to compensate

Correct. A fast heart and cool, pale skin come before any pressure change.

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Joyce University of Nursing and Health Sciences

Course resources by **Dr. Vidya Garg**